NOT MEASUREMENT SENSITIVE

A-A-59686
May 25, 2001

COMMERCIAL ITEM DESCRIPTION

PREHEATER, ENGINE COOLANT, DIESEL-FIRED, 24-VOLT DC

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers requirements for diesel-fired engine coolant preheaters, known hereinafter as preheaters. The purpose of a preheater is to preheat water-cooled diesel engines that are at ambient temperatures from -18 degrees Celsius (°C) to -46°C (0 degrees Fahrenheit (°F) to -50°F) so they can be started without damaging the engine. Preheaters bought in accordance with (IAW) this CID should not be used on systems with operational ether start. Type 1 and type 2 preheaters are recommended for engine displacements of up to 8 liters (L) (488 cubic inches (in^3)); Type 3 preheaters are recommended for engine displacements of 9 to 12.5 L (549 to 763 in^3).

2. CLASSIFICATION. Preheaters shall be of the following types:

Type 1 - 2.2 to 5.0 kilowatts (kW) (7,500 to 17,000 British thermal units per hour (Btu/hr)) capacity, with enclosed heater/water pump assembly.

Type 2 - 2.2 to 5.0 kW capacity, with heater and water pump supplied separately, to accommodate space limitations on host vehicles.

Type 3 - 1.5 to 9.0 kW (5,000 to 31,000 Btu/hr) capacity, with enclosed heater/water pump assembly.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/IE, Warren, MI 48397-5000.

AMSC N/A FSC 2990

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3. SALIENT CHARACTERISTICS.

3.1 Materials. Unless specified herein, materials shall be IAW the manufacturer's material specifications. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 4).

3.1.1 Dissimilar metals. Dissimilar metals shall electrically insulated from one another to prevent galvanic corrosion.

3.1.2 Fuel resistance. Preheater components that are exposed to fuel shall be resistant to damage caused by direct contact with the fuel (see 3.2.7.1).

3.1.3 Corrosion resistance. Preheater components shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide against corrosion and deterioration during storage and operation.

3.1.4 Hazardous materials. Asbestos and cadmium-plated components shall not be used in the preheater or any of its components.

3.2 Design and construction.

3.2.1 Type 1 preheaters. Type 1 preheaters shall have a minimum of two heat settings (high/low) and shall have a heat output capacity of 2.2 to 5.0 kW. Type 1 preheaters shall be provided with a heater/water pump assembly enclosed in a weather resistant enclosure or box with a quick release cover for easy access (see 3.2.6). Type 1 preheaters shall have all interface connections within the enclosure or surface mounted on the exterior of the enclosure. Type 1 preheaters shall be provided with the components specified in table I.

3.2.2 Type 2 preheaters. Type 2 preheaters shall have a minimum of two heat settings (high/low) and shall have a heat output capacity of 2.2 to 5.0 kW. Type 2 preheaters shall be provided as a "kit", with individual components supplied for specialized installation within a host vehicle, where space is limited. The kit shall contain all necessary components for a complete installation into the host vehicle, including, but not limited to those specified in table I.

3.2.3 Type 3 preheaters. Type 3 preheaters shall have a minimum of four heat settings (boost/high/medium/low) or variable heat output, and shall have a heat output capacity of 1.5 to 9.0 kW (see 7.5). Type 3 preheaters shall be provided with a heater/water pump assembly enclosed in a weather resistant enclosure or box with a quick release cover for easy access (see 3.2.6). Type 3 preheaters shall have all interface connections within the enclosure or surface mounted on the exterior of the enclosure. Type 3 preheaters shall be provided with the components specified in table I.

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TABLE I Preheater component list.

Component	Type 1	Type 2	Type 3
Heater/water pump, enclosed	X		X
Heater/water pump, separate		X	
Fuel metering pump	X	X	X
Fuel hose, pressure, 6.1 meters (m) (20 feet (ft.)), with clamps	X	X	X
Fuel hose, suction, 1.8 m (6 ft.), with clamps	X	X	X
Control unit/timer and harness (see 3.2.8 and 3.2.9)	X	X	X
Electrical harnesses, electrical connectors and wiring		X	
Exhaust pipe	X	X	X
Fuel tank pick-up pipe and connections	X	X	X
Mounting brackets/fasteners	X	X	X
Coolant hoses and connectors		X	
Data plates	X	X	X
Operator/maintenance/installation manuals	X	X	X

3.2.4 Power requirements. Preheaters shall start and operate at applied voltages between 20 - 28.5 volts (V) direct current (dc). Preheaters shall cycle for up to two hours after each start-up unless the operator manually turns it off or a fault occurs. Preheaters shall shut off whenever low voltage (20 V or less) is detected. Maximum electrical power consumption with water pump at 24 V shall be as specified in table II.

TABLE II. Power consumption.

Preheater type	Power, watts (W)		
	Boost setting	High setting	Low setting
1 and 2	-	55	35
3	125	85	37

3.2.5 Weight and size. Preheater and weather-resistant enclosure shall not exceed the weights and sizes specified in table III.

TABLE III. Weight and size restrictions.

Preheater type	Weight, kilograms (kg) (pounds (lb))	Size, millimeter (mm) (inch (in.))
1	10.0 (22)	150 x 320 x 240 (6.0 x 12.5 x 9.5)
2 1/	5.0 (11)	250 x 110 x 200 (10.0 x 4.5 x 8.0)
3	21.0 (46)	430 x 300 x 230 (17.0 x 12.0 x 9.0)

1/ As type 2 preheaters have no enclosure, the restrictions are for the heater (see 3.2.2).

3.2.6 Preheater enclosure. Type 1 and type 3 preheaters (see 3.2.1 and 3.2.3) shall come with an enclosure that can be secured (bolted shut or locked with a padlock). The enclosure shall protect preheater from outside environment (rain, humidity, dust, snow, salt, engine fluids).

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3.2.7 Interfaces and compatibility.

3.2.7.1 Fuel compatibility. Preheaters shall operate on and be compatible with diesel fuels IAW A-A-52557, MIL-DTL-83133 (JP-8), ASTM-D1655 (Jet A-1) and MIL-DTL-5624 (JP-4).

3.2.7.2 Coolant compatibility. Preheaters shall be compatible with ethylene glycol (Type IP) and propylene glycol (Type II) based antifreezes IAW A-A-52624. Preheaters shall also be compatible with antifreeze having corrosion inhibitors IAW MIL-A-53009.

3.2.7.3 Electrical interface. Preheaters shall come with all required wiring harnesses, fuses, relays and mounting hardware, and shall have quick-disconnect waterproof connectors. All wires shall be distinguishable from each other and match those in the schematic and electrical diagram provided in the preheater manual and installation instructions.

3.2.7.3.1 Type 1 and type 3. The electrical power harness and preheater control unit (on/off) harness shall terminate at MS connector MS3102E20-16P (see table IV), surface mounted on the exterior of the enclosure for connection with the vehicle harness (power, control, diagnostic).

TABLE IV. MS3102 insert arraignment.

Pin designation	Gage	Circuit
A	16	Fuel pump positive (+)
B	16	Fuel pump negative (-)
C	16	Control (+)
D	16	Control (-)
E	16	Heater enable
F	16	Heater diagnostics
G	16	Free terminal
H	12	Battery (+)
I	12	Battery (-)

3.2.7.3.2 Type 2. The electrical power harness and preheater control unit (on/off) harness shall be 3.0 m (10 ft.) in length, shall terminate with MS3102E20-16P, and shall be capable of being shortened during installation. Pins on all connectors shall be clearly labeled and coded for interchangeability between manufacturers.

3.2.7.4 Water/coolant interface.

3.2.7.4.1 Type 1 and type 3. The preheater coolant inlet/outlet ports shall be 3/4 in. outside diameter (O.D.), labeled and extended through the enclosure.

3.2.7.4.2 Type 2. The preheater coolant inlet/outlet ports shall be 3/4 in. O.D, and shall be provided with all the required clamps for connections at the inlet, outlet and water pump.

3.2.7.5 Air intake/exhaust interface. Preheater intake and exhaust ports shall accept combustion air intake and exhaust pipes. Preheaters shall come with a flexible 1.8 m long exhaust pipe, which can be shortened. All required clamps shall be included.

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3.2.7.6 Fuel interface. A self-priming fuel pump for pumping fuel from the host engine fuel tank to the preheater shall be included with the preheater. Fuel pump installation instructions shall clearly identify the method of mounting the fuel pump relative to the heater, fuel tank and vertical and horizontal planes. Fuel pump electrical harness shall be 152 mm (6 in.) in length, and terminate in MS3100E10SL-4P, for connection with the vehicle harness. A fuel tank standpipe or pickup pipe shall be provided for withdrawing fuel from the host vehicle fuel tank. Preheaters shall come with all connecting hardware, including but not limited to, hose clamps and 6.1 m of fuel pressure line. Preheaters shall be provided with 1.8 m of suction line.

3.2.8 Preheater control unit. Preheaters shall have a control unit to enable the operator to manually turn the preheater on and off. A light next to or part of the control unit shall clearly identify to the operator whether or not the preheater is working when the switch is set to the on position. The control unit shall have provisions for instrument panel mounting and shall be fitted with MS3100E14S-2P, for connection with vehicle harness. For type 3 preheaters, the control unit shall also function as a fault code retrieval device for the preheater diagnostics specified in 3.2.10.

3.2.9 Timer. When specified (see 7.3), preheaters shall be equipped with a programmable 7-day, 24-hour timer in place of the control unit (see 3.2.8). The timer shall function as the control unit and allow the user to preset up to 3 different preheater start and stop times per day. The preset times shall require activation of the preset timer after each use of preheater. The timer shall also function as a fault code retrieval device for the preheater diagnostics specified in 3.2.10. The timer shall have provisions for instrument panel mounting and shall be fitted with MS3100E14S-2P, for connection with vehicle harness.

3.2.10 Preheater diagnostics. When equipped with the optional timer (see 3.2.9), preheaters shall have diagnostic capability to diagnose operational faults. When not equipped with the optional timer, only type 3 preheaters shall have diagnostic capability. At a minimum, the diagnostic capabilities provided shall include low battery, no coolant, low coolant, no fuel, overheat, open circuit, short circuit, defective fuel pump, defective coolant pump, bad flame and bad glowplug. If the preheater fails to start after three attempts, it shall automatically exit the start mode, diagnose the fault and indicate the reason for the failed start within five minutes of initial start attempt. Diagnostic fault codes shall be in the form of distinguishable audio or visual signals. A decal or data plate with fault code definitions shall be mounted in a visible location inside of the heater enclosure on type 1 and type 3 preheaters. Type 2 preheaters shall come with a fault code definition decal for users to mount.

3.2.11 Overheat protection. Preheaters shall have an overheat switch that stops operation of the preheater in cases of low coolant, no coolant and overheating (coolant temperature in excess of 93°C (200°F)).

3.3 Performance. After each start the preheater shall bring the coolant temperature at the heater water pump inlet up to 72°C (162°F) and maintain it between 72°C and 85°C (185°F) during operation.

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3.3.1 Engine cold start. Preheaters shall be able to preheat a cast iron diesel engine, cold-soaked to -46°C, to a temperature of 32 ± 6°C (90 ± 10°F) at the water pump inlet, within one hour after the preheater is turned on.

3.3.2 Fuel consumption. Preheater fuel consumption shall not be greater than as specified in table V.

TABLE V. Fuel consumption.

Preheater type	Fuel consumption, liters per hour (L/hr) (gallons per hour (gal/hr))		
	Boost setting	High setting	Low setting
1 and 2	-	0.62 (0.14)	0.30 (0.07)
3	1.20 (0.26)	0.91 (0.20)	0.24 (0.05)

3.3.3 Environmental conditions. Preheaters and all components, including optional timer, shall reliably start and operate at ambient temperatures from -46 to 21°C (-50 to 70°F) within four minutes after activation by operator or by timer. Preheater and control unit shall be able to withstand extended storage at temperatures ranging from -46 to 49°C (-50 to 120°F).

3.3.4 Noise. Preheater steady state noise shall not exceed 85 decibels (dB(A)) at a radius of 2.44 m (8 ft.).

3.3.5 Vibration. Preheaters shall be able to withstand shock and vibrations of at least 3 gravitational units (g) without breakage, loosening of parts, leakage, abrasion, or other physical damage.

3.3.6 Electromagnetic compatibility. Electromagnetic energy from wiring harnesses/electrical system shall not interface with operation of electrical systems on board or near vehicles. When specified (see 7.3), preheaters shall be IAW MIL-STD-461.

3.3.7 Slope and grade. Preheaters shall function and shall not leak or sustain damaged when the host vehicle is operating on grades of up to 60% up hill or down hill and side slopes of up to 40%. Preheaters shall be able to operate without malfunctioning when mounted on a vehicle parked on 15% slope in any direction.

3.3.8 Smoke density. Within 7 minutes after preheater has started, its exhaust gases shall not exceed a smoke spot number of three IAW ASTM-D2156.

3.4 Color. The color of the preheater enclosure shall be Green 383, IAW Drawing 12369003.

3.5 Safety. All preheater components subject to high operating temperatures or energized electrically shall be insulated, enclosed or guarded. All preheaters shall comply with Department of Transportation (DOT) standards in effect on date of manufacture.

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3.6 Manuals and installation instructions. Preheaters shall come with an operator manual and a maintenance manual. Operator manual shall include instruction for operating a basic preheater plus the optional timer. Maintenance manual shall include a parts list of all repairable or replaceable parts and instructions for diagnosing and repairing defective preheaters. Preheaters shall come with complete general installation instructions that include mating to the engine, electrical system, cooling system and fuel system of host vehicle. Electrical schematics in manuals and installations instructions shall clearly identify each wire along with its function. Pins on all connectors shall be clearly identified in the manuals and installation instructions and match what is on the actual connector.

3.7 Data plates. Preheaters shall have a data plate with permanent and legible markings, identifying, as a minimum, the manufacturer's identification code (CAGE), the manufacturers model number and preheater serial number, the contract number, and the Part or Identification Number (PIN) (see 7.1).

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, IAW paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

6. PACKAGING. Unless otherwise specified (see 7.3), the preservation, packaging, packing, labeling and marking of preheater kits shall be developed by the contractor IAW MIL-STD-2073/1, Appendix D, Method 41, unit pack quantity of 1 kit each, Level M/B.

6.1 Unit pack container testing. The unit pack shall be subject to tests IAW MIL-STD-2073/1, Appendix F, paragraphs F.3.1, F.4.1, F.4.2, and F.5.1. The unit pack design and engineering data shall be submitted to the DoD activity (TACOM Packaging) for approval.

6.2 Unit pack quality assurance. The contractor is responsible for unit pack examinations, inspections, and tests necessary to ensure the acceptability of the package.

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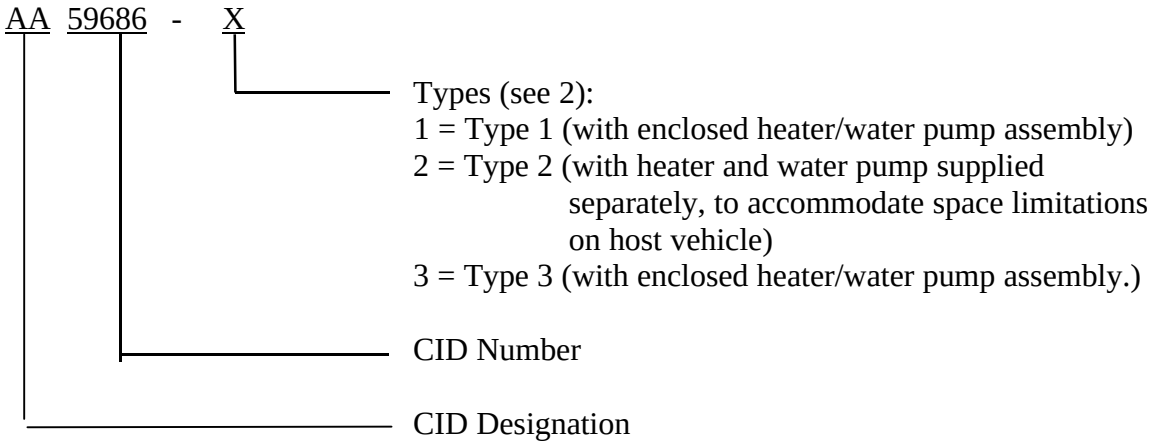
7. NOTES.

7.1 Part or Identification Number (PIN). The following PIN procedure is for Government purposes and does not constitute a requirement for the contractor. The PINs to be used for preheaters acquired to this CID are created as follows:



7.2 Source of documents.

7.2.1 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.2.2 Copies of A-A-52557 “Fuel Oil, Diesel; For Posts, Camps And Stations”; A-A-52624 “Antifreeze, Multi Engine Type”; MIL-A-53009 “Additive, Antifreeze Extender, Liquid Cooling Systems”; MIL-DTL-5624 “Turbine Fuel, Aviation, Grades JP-4, JP-5, AND JP-5/JP-8 ST”; MIL-DTL-83133 “Turbine Fuels, Aviation, Kerosene Types, NATO F-34(JP-8), NATO F-35, And JP-8 + 100”; MIL-STD-2073/1 “DOD Standard Practice For Military Packaging”; MIL-STD-461 “Requirements For The Control Of Electromagnetic Interference Characteristics Of Subsystems And Equipment”; MS3100 “Connector, Receptacle, Electric, Wall Mounting, Solder Contacts, AN Type”; and MS3102 “Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type” are available from the Document Automation and Production Service, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or at: http://assist.daps.mil.

7.2.3 Copies of Drawing 12369003 are available from the US Tank-automotive and Armaments Command (TACOM), Warren, MI 48397-5000.

7.2.4 Copies of ASTM-D1655 “Aviation Turbine Fuels” and ASTM-D2156 “Smoke Density In Flue Gases From Burning Distillate Fuels” are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or at: http://www.astm.org/.

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7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. If a timer is to be provided (see 3.2.9).
c. If preheater is to be IAW MIL-STD-461 (see 3.3.6).
d. Product conformance provisions.
e. If packaging requirements are other than as specified (see 6).

7.4 Larger capacity preheaters. A larger preheater should be considered if the preheater will also be used to heat hydraulic reservoirs, engine oil, transmission oil, or provide supplemental cab heat.

7.5 Boost heat setting. The boost heat setting is to quickly provide maximum heat during start-up to bring the preheater and coolant up to operating temperature.

7.6 Vehicle preparation kit for type 1 and type 3 preheaters. Host vehicles to be outfitted with type 1 and type 3 preheaters typically require the preheater vehicle preparation kit which is peculiar to each vehicle.

7.7 Commercial products. Listed below are manufacturers that offer coolant preheaters. They are listed for information purpose only.

a. Espar Heater Systems, (800) 387-4800.
b. Webasto Thermosystems, (800) 432-8371.

7.8 Key words.

Cold start aid
Heater

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army - AT

(Project 2990-0126)

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